



DataLift Business Enablement & Services LLC

05/20/2026

BUILDING AN END TO END RETAIL ANALYTICS PLATFORM

**From raw CSV files to executive dashboard reporting
using a Medallion Lakehouse Architecture**



NAME OF PROJECT:
**Retail Sales & Inventory
Analysis**

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Agenda

03	Business Problem
04	Architecture Overview
05	Bronze Layer
06	Silver Layer
07	Gold Layer
08	Executive Sales & Retail Performance Dashboard
09	Product Performance & Profitability Dashboard
10	Inventory Health & Stock Monitoring Dashboard
11	Key Takeaways

Business Problem

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Problem Statement

Retail organizations often struggle with:

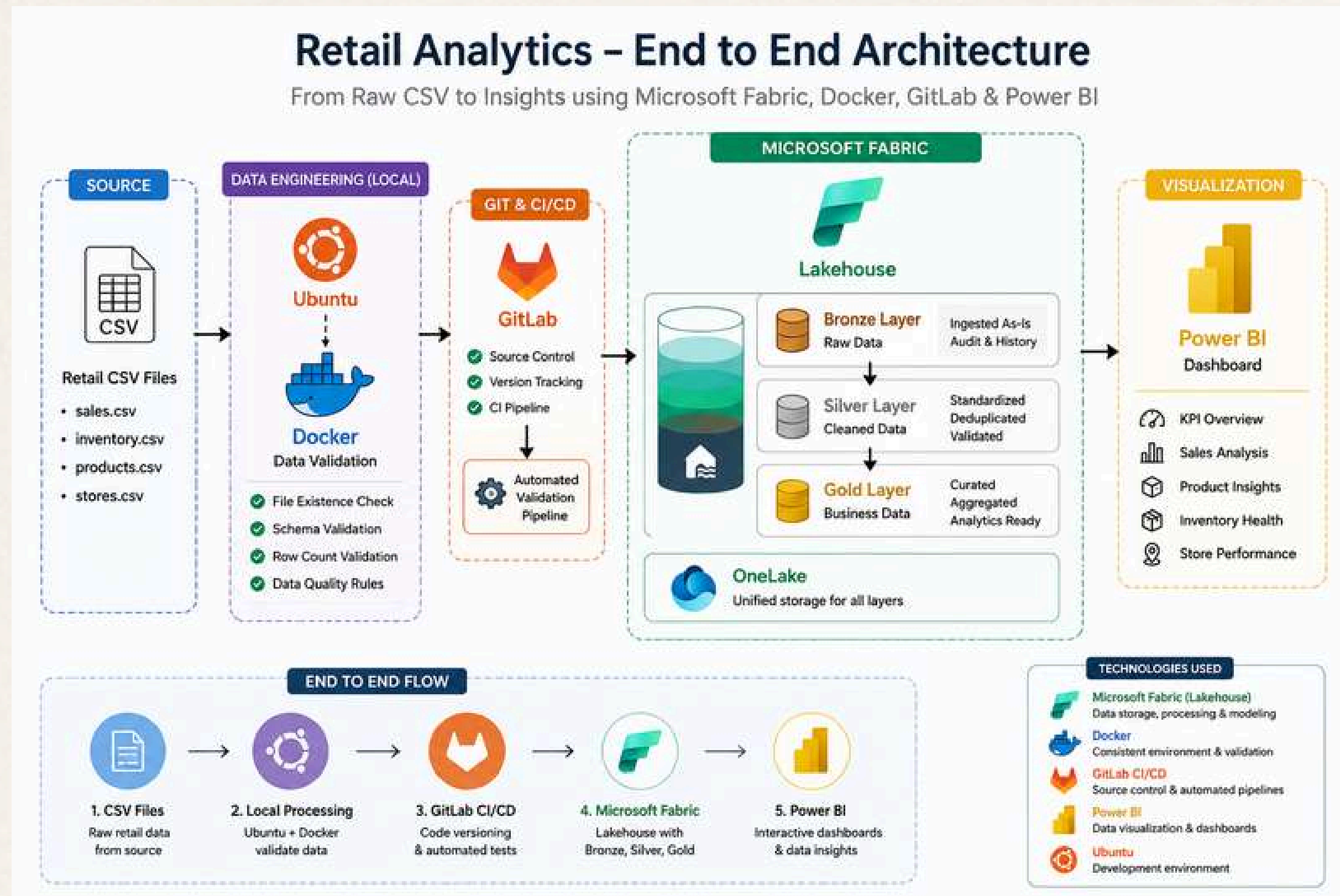
- siloed sales and inventory data
- inconsistent reporting
- lack of inventory visibility
- delayed analytics workflows

This project simulates a modern cloud data engineering solution that centralizes ingestion, cleansing, modeling, and reporting.



Architecture Diagram

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Bronze Layer

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Raw Data Ingestion

Completion Process:

- uploaded CSV files into Fabric Lakehouse
- raw ingestion strategy
- validation checks
- medallion architecture

BRONZE LAYER

Raw data ingestion in Microsoft Fabric Lakehouse

Data is ingested from CSV files into the Bronze layer as-is (raw format) for audit, traceability and historical record.

```
graph LR; A[CSV Files] --> B[Bronze]; B --> C[Silver]; C --> D[Gold]
```

WHAT IS THE BRONZE LAYER?

- ✓ Raw data is landed as-is from source CSV files.
- ✓ No transformations are applied.
- ✓ Schema is preserved exactly as received.
- ✓ Used for audit, reprocessing and historical traceability.

SOURCE FILES (CSV)

	sales.csv Sales transactions
	inventory.csv Inventory snapshots
	products.csv Product master data
	stores.csv Store master data

MICROSOFT FABRIC LAKEHOUSE

Lakehouse Explorer

- ▼ Retail_Lakehouse
 - > Tables
 - ▼ bronze
 - bronze_sales
 - bronze_inventory
 - bronze_products
 - bronze_stores
 - > silver
 - > gold
 - > Files
 - > Schemas

BRONZE LAYER TABLES (RAW DATA)

✓ INGESTED

Table Name	Description	Row Count	File Source
bronze_sales	Raw sales transactions	10,000	sales.csv
bronze_inventory	Raw inventory snapshots	5,000	inventory.csv
bronze_products	Raw product master	100	products.csv
bronze_stores	Raw store master	50	stores.csv

SAMPLE PREVIEW – bronze_sales

sale_id	date_key	store_id	product_id	qty	unit_price	discount	total
1	20240101	S001	P001	2	19.99	0.00	39.98
2	20240101	S002	P005	1	9.99	0.00	9.99
3	20240101	S001	P002	3	14.99	1.50	43.47
4	20240102	S003	P003	2	7.99	0.00	15.98
5	20240102	S002	P001	1	19.99	0.00	19.99
...	...	...	...	...	...	...	...

KEY TAKEAWAYS

- Raw & Immutable**
Data is stored in its original format.
- Audit & Traceability**
Full historical record for reprocessing.
- Foundation Layer**
Serves as the source for Silver layer.
- No Transformations**
Data is not cleansed or modified here.

💡 Bronze layer ensures data reliability, transparency and provides a single source of truth for the pipeline.

Silver Layer

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Data Cleansing & Standardization

Completion Process:

- PySpark notebook transformations
- schema enforcement
- deduplication
- derived metrics
- quality validation

SILVER LAYER

Cleaned & standardized data in Microsoft Fabric Lakehouse

Data is cleansed, deduplicated, standardized and validated to ensure consistency and quality across the business.



WHAT IS THE SILVER LAYER?

- ✓ Data is cleansed, deduplicated and standardized.
- ✓ Data quality checks are applied.
- ✓ Conformed to business rules and naming standards.
- ✓ Ready for analytics and business processing.

SOURCE FILES (BRONZE LAYER)

	bronze_sales	Raw sales transactions
	bronze_inventory	Raw inventory snapshots
	bronze_products	Raw product master
	bronze_stores	Raw store master

Microsoft Fabric Lakehouse

Lakehouse Explorer

- Retail_Lakehouse
 - Tables
 - bronze
 - silver
 - silver_sales
 - silver_inventory
 - silver_products
 - silver_stores
 - gold
 - Files
 - Schemas

SILVER LAYER TABLES (CLEANED DATA)

✓ CLEANED

Table Name	Description	Row Count	Source Table (Bronze)
silver_sales	Cleaned sales transactions	9,842	bronze_sales
silver_inventory	Cleaned inventory snapshots	4,912	bronze_inventory
silver_products	Cleaned product master	98	bronze_products
silver_stores	Cleaned store master	50	bronze_stores

SAMPLE PREVIEW – silver_sales

sale_id	date_key	store_id	product_id	qty	unit_price	discount	total
1	20240101	S001	P001	2	19.99	0.00	39.98
2	20240101	S002	P005	1	9.99	0.00	9.99
3	20240101	S001	P002	3	14.99	0.00	44.97
4	20240102	S003	P003	2	7.99	0.00	15.98
5	20240102	S002	P001	1	19.99	0.00	19.99
...	...	...	...	...	...	...	...

KEY TAKEAWAYS

-  **Clean & Consistent**
Data is cleansed, standardized and consistent.
-  **Quality Assured**
Data quality checks ensure accuracy and reliability.
-  **Business Ready**
Conformed to business rules and naming standards.
-  **Ready for Analytics**
Data is prepared for analytics and reporting.

 Silver layer ensures high-quality, consistent and reliable data for business processing and analytics.

Gold Layer

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Analytics Ready Modeling

Completion Process:

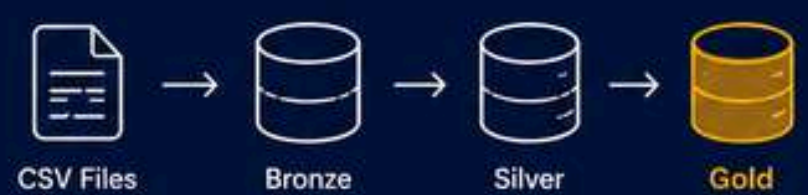
- fact and dimension modeling
- semantic layer preparation
- KPI optimization

Designed for Power BI semantic modeling and executive reporting

GOLD LAYER

Curated business data in Microsoft Fabric Lakehouse

Data is modeled, aggregated and enriched to deliver business-ready datasets for analytics and reporting.



WHAT IS THE GOLD LAYER?

- ✓ Data is modeled, aggregated and enriched.
- ✓ Business logic and calculations are applied.
- ✓ Optimized for reporting and analytics.
- ✓ Conformed dimensions and fact tables.
- ✓ Trusted, consistent single source of truth for the business.

Microsoft Fabric Lakehouse

Lakehouse Explorer

- Retail_Lakehouse
 - Tables
 - bronze
 - silver
 - gold
 - gold_fact_sales
 - gold_fact_inventory
 - gold_dim_product
 - gold_dim_store
 - gold_dim_date
 - Files
 - Schemas

GOLD LAYER TABLES (BUSINESS READY)

✓ CURATED

Table Name	Description	Row Count	Source (Silver)
gold_fact_sales	Sales fact table (aggregated)	8,754	silver_sales
gold_fact_inventory	Inventory fact table (daily)	4,912	silver_inventory
gold_dim_product	Product dimension	98	silver_products
gold_dim_store	Store dimension	50	silver_stores
gold_dim_date	Date dimension	1,825	—

SAMPLE PREVIEW – gold_fact_sales

date_key	store_id	product_id	total_qty	total_revenue	total_profit	order_count
20240101	S001	P001	12	239.88	78.45	1
20240101	S002	P005	8	79.92	28.10	1
20240101	S001	P002	15	224.85	73.20	1
20240102	S003	P003	6	47.94	15.30	1
20240102	S002	P001	9	179.91	58.70	1
...	...	...	...	...	...	...

KEY TAKEAWAYS

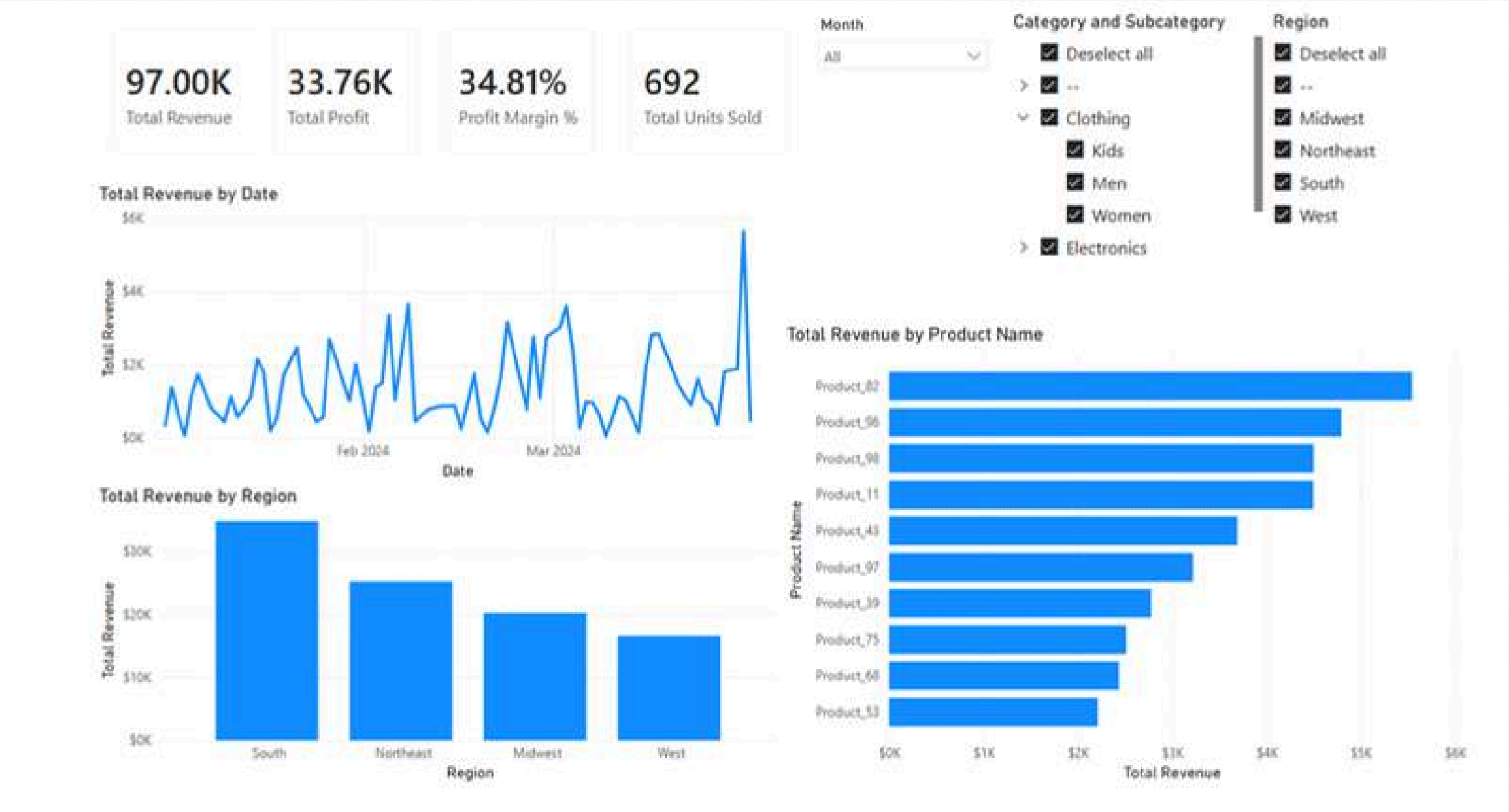
- Business Ready**
Data is curated and modeled for business use.
- Aggregated & Optimized**
Aggregations and business logic applied for performance.
- Single Source of Truth**
Trusted and consistent data for reporting and insights.
- Reporting Ready**
Optimized for dashboards, KPIs and self-service BI.

Gold layer delivers trusted, business-ready data that drives accurate reporting, KPIs and data-driven decisions.

Executive Sales & Retail Performance Dashboard

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Interactive Power BI dashboard built using Microsoft Fabric Gold Layer tables and semantic modeling.

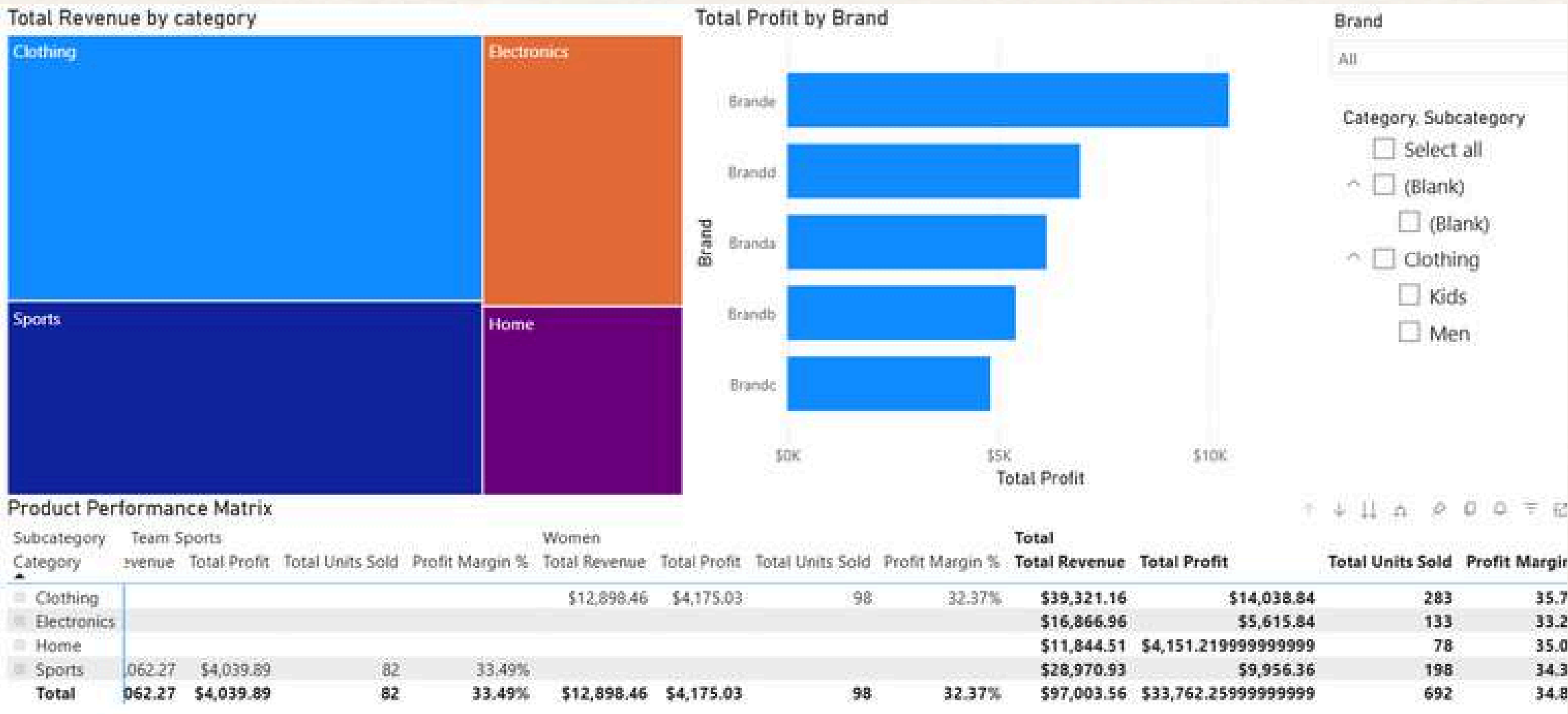


- Developed a Power BI executive dashboard connected to Microsoft Fabric Lakehouse Gold tables.
- Modeled semantic relationships between fact and dimension tables for enterprise-style reporting.
- Built KPI cards, trend analysis, regional performance metrics, and product ranking visualizations.
- Enabled self-service analytics using interactive slicers and dynamic filtering capabilities.
- Applied medallion architecture principles (Bronze → Silver → Gold) to support trusted reporting.
- Integrated curated business-ready datasets optimized for analytics consumption.

Product Performance & Profitability Dashboard

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Interactive product analytics dashboard built using Microsoft Fabric Gold Layer sales tables and semantic modeling.



- Built a product performance dashboard using Microsoft Fabric and Power BI semantic modeling.
- Designed analytics for category-level revenue, brand profitability, and product performance tracking.
- Enabled interactive drill-down analysis across categories, subcategories, and brands.
- Identified top-performing product categories and highest profit-generating brands through dynamic reporting.
- Leveraged Gold layer fact and dimension tables to support scalable business analytics.
- Delivered centralized product insights for revenue optimization and executive decision-making.
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Inventory Health & Stock Monitoring Dashboard

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Interactive inventory monitoring dashboard built using Microsoft Fabric Gold Layer inventory tables and semantic modeling.



- Built an inventory health dashboard using Microsoft Fabric and Power BI semantic modeling.
- Designed KPI tracking for total inventory on hand and low stock item monitoring.
- Enabled operational visibility into inventory distribution across stores and product categories.
- Identified low stock products requiring replenishment through dynamic filtering and inventory alerts.
- Leveraged Gold layer inventory fact tables for optimized reporting and analytics performance.
- Supported inventory management decision-making through centralized business-ready reporting.

Key Takeaways

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Summary

- End-to-end cloud data engineering using Microsoft Fabric and medallion architecture.
- Centralized ingestion, cleansing, and modeling for trusted analytics.
- Executive dashboards built on semantic models for self-service reporting.
- Scalable, repeatable patterns applicable to real-world retail environments.
- Data-driven decision-making enabled through modern tooling and best practices.

